

The Conative Bridge: Active Inference Meets Integrated Information Theory (Φ)

Uniting the Free Energy Principle, IIT 4.0, and The 6th Axiom of Consciousness (The Will to Exist)

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Framework: The Conative-Integrative Framework (CIF)
Date: September 2026 • Working Draft

★ THE FUNDAMENTAL MASTER EQUIVALENCE OF CONSCIOUS AUTOPOIESIS ★

$$\pi^* = \arg \min_{\pi} \sum_{\tau} G(\pi, \tau) \iff \mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\Phi > 0)$$

3rd-Person Physics (Karl Friston): Expected Free Energy Minimization across the Markov Blanket $\equiv$ 1st-Person Phenomenology (Giulio Tononi & Thomas Riebl): Autopoietic Persistence of Causal Power via the 6th Axiom (The Will to Exist)

⚡ 1. 3rd-Person Exteriority (FEP)

Karl Friston (Active Inference & POMDP): Self-organizing biological systems preserve physical existence against thermodynamic decay by minimizing **Variational Free Energy (F)** and selecting policy π^* to minimize **Expected Free Energy (G)**:

$$F \geq -\ln p(o) \quad \mid \quad G(\pi) = \text{Pragmatic} + \text{Epistemic}$$

- **Perceptual Inference:** Updating hidden states $q(s)$ via sensory prediction error $\gamma(o - g(s))$.
- **Active Inference:** Executing actions u to force incoming sensory states o toward prior homeostatic preferences (C).
- **Markov Blanket:** Statistical phase-space partition separating Internal States (μ) from External States (η).

🔥 2. The 6th Axiom of IIT (Riebl)

The Missing Conative Axiom: Standard IIT 4.0 defines five static axioms (Existence, Intrinsicity, Information, Integration, Exclusion), creating the *Paradox of Transient Causal Phantoms* (static lifeless logic gates having high Φ).

$$\mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\text{Autopoietic Persistence})$$

- **Conative Dynamics:** Genuine consciousness requires active temporal self-preservation (Spinoza's *Conatus*, Schopenhauer's *Wille zum Leben*).
- **Action-Perception Loop:** Φ is sustained because the system actively executes policies π^* that prevent phase-space collapse ($\Phi \rightarrow 0$).
- **The Edge of Chaos:** Critical metastability balancing differentiation and integration to maximize dynamic Φ .

💡 3. 1st-Person Interiority (IIT 4.0)

Giulio Tononi (Expanded IIT 4.0): The qualitative interiority of an entity is strictly identical to its **Intrinsic Integrated Causal Structure (S)** and causal power ($\Phi > 0$) across the Minimum Information Partition (MIP):

$$\Phi(M_1; M_2) = \frac{1}{2} (\ln \det \Sigma_{M_1} + \ln \det \Sigma_{M_2} - \ln \det \Sigma_{\text{Whole}})$$

- **Maximal Substrate:** Consciousness exists only at the spatiotemporal grain where integrated cause-effect power Φ reaches a local maximum.
- **Causal Irreducibility:** A system is conscious if and only if its whole exerts causal power over itself above and beyond its independent parts ($\Phi > 0$).

* **Formal Scientific Footnote on the 6th Axiom:** The 6th Axiom ("The Will to Exist / Autopoietic Causal Persistence": $\mathbb{E}[\Phi(t+1) \mid \pi] \geq \Phi(t)$) is Thomas Riebl's proposed theoretical extension to Integrated Information Theory 4.0 (IIT 4.0 by Tononi et al.). While standard IIT 4.0 formulates five static phenomenological axioms (Existence, Intrinsicity, Information, Integration, Exclusion), Riebl's 6th Axiom introduces the necessary *temporal, autopoietic action-perception requirement*. It resolves the "Transient Causal Phantom" paradox and mathematically bridges the static causal topology of IIT with Karl Friston's dynamic Free Energy Principle, establishing that conscious entities are self-sustaining active inference networks driven by an intrinsic Will to Exist.